

Ideation Phase Empathize & Discover

Date	12/03/2026
Team ID	SWUID20264931
Project Name	Electric vehicle Charge and Range
Maximum Marks	

Empathy Map Canvas

Topic: EV Charge and Range

1. User (Who are we empathizing with?)

Electric Vehicle Owner / Potential EV Buyer

People who use EVs for **daily commuting, office travel, and long-distance trips** and want a reliable and convenient charging experience.

2. Says (What the user says)

- “I hope my EV battery lasts for the whole trip.”
- “Charging takes too long compared to filling petrol.”
- “I wish there were more charging stations on highways.”
- “I need a quick way to check my vehicle’s remaining range.”
- “Fast charging would make EVs much more convenient.”

3. Thinks (What the user thinks)

- “What if my battery runs out before I find a charging station?”
- “Is my EV suitable for long-distance travel?”
- “Charging infrastructure should be more widely available.”
- “I need better battery technology for longer range.”
- “Planning trips with an EV requires too much effort.”

4. Does (What the user does)

- Checks battery level frequently while driving.
- Uses navigation apps to locate nearby charging stations.
- Plans routes carefully to avoid running out of charge.
- Charges the vehicle overnight at home.
- Searches online for fast charging options.

5. Feels (What the user feels)

- **Anxious** about battery range during long trips.
- **Frustrated** when charging stations are unavailable.
- **Concerned** about charging time.
- **Satisfied** when fast and convenient charging is available.
- **Motivated** to use EVs for environmental benefits.

6. Pain Points (User Challenges)

- Limited EV driving range.
- Long charging time.
- Limited number of charging stations.
- Difficulty finding available charging points.
- High cost of battery replacement.

7. Gains (User Goals / Expectations)

- Faster EV charging technology.
- Longer battery range.
- Easily accessible charging stations.
- Smart apps to monitor battery and range.
- Reliable EV infrastructure for long-distance travel.

Persona: Electric Vehicle Charging Experience

